

EXPLAINED EXAMPLE 7 – ISENTROPIC EXPANSION

- ① Steam expands isentropically from a starting temperature of 500 K and pressure of 2 MPa to a saturated vapor. Determine the final temperature and pressure of the fluid at the end of the process, and draw this process on a T-s diagram. What would the final temperature be if air, starting at the same temperature and pressure, ended at the same final pressure? Assume the following properties for air: $c_p=1000 \text{ J/kg-K}$ and $c_v=714.29 \text{ J/kg-K}$.

Given: $T_1 = 500 \text{ K}$
 ② $P_1 = 2 \text{ MPa}$
 $x_2 = 1$
 $S_2 = S_1$

Find: T_2, P_2 steam
 T_2, P_2 air

Assume: isentropic
 ① $\text{H}_2\text{O} = \text{S.C.S}$
 air = ideal
 quasi-steady

→ Steam State 1: $T_1 = 500 \text{ K}$
 ② $P_1 = 2 \text{ MPa}$ } Table D.2 $P_1 = 2 \text{ MPa}$
 $T_1 > T_{\text{sat}} \rightarrow \text{vapor}$

→ Table D.3K → $S_1 = 6.4265 \frac{\text{kJ}}{\text{K}}$

→ Steam State 2: $x_2 = 1$
 ① $S_2 = S_1 = 6.4265 \frac{\text{kJ}}{\text{K}}$

→ Table D.1: $S_2 = S_g \rightarrow$ in between $T_a = 470 \text{ K}$
 $T_b = 475 \text{ K}$

Linear
 → Interpolation: $T_2 = \left(\frac{S_2 - S_a}{S_b - S_a} \right) (T_b - T_a) + T_a$

$$= \left(\frac{6.4265 - 6.4538}{6.4164 - 6.4538} \right) (475 - 470) + 470$$

$$\boxed{T_2 = 473.65 \text{ K}}$$

$P_{sat}: P_a = 1.455 \text{ MPa} \quad P_b = 1.616 \text{ MPa}$

② → linear interpolation

$$P_2 = \left(\frac{s_2 - s_a}{s_b - s_a} \right) (P_b - P_a) + P_a$$

$$= \left(\frac{6.4265 - 6.4538}{6.4164 - 6.4538} \right) (1.616 - 1.455) + 1.455$$

$$P_2 = 1.573 \text{ MPa}$$



→ air: isentropic relations

②

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

where $\gamma = \frac{C_p}{C_v} = \frac{1000}{714.29} = 1.4$

$$= 500 \left(\frac{1.573}{2} \right)^{\frac{0.4}{1.4}}$$

$$\Rightarrow T_2 = 466.84 \text{ K}$$